EECS 484- Database Management Systems

Hash-Based Indexes

Chapter 11

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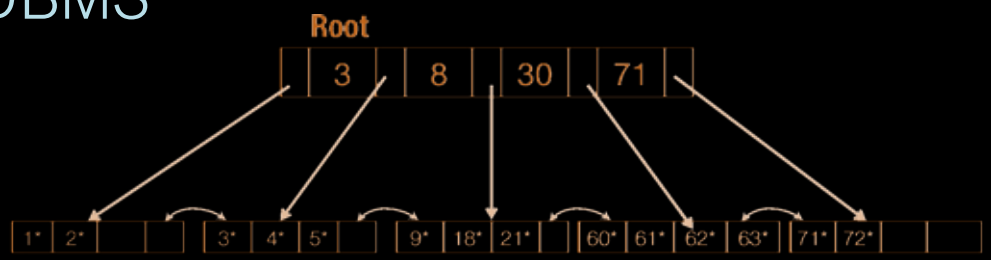
Recap: B+ Trees

- Height-balanced tree search structure
- Ideal for range queries, good for equality queries
- Insertion / deletion / search costs: O($\log_F N$)
- Invariants to maintain when inserting/deleting data entries:
  - Search-tree property
  - All nodes must be at least ½ full (except root node)
  - Root node is allowed to have a single entry
- Most widely used index in DBMS



Today's Learning Goals

1. What is a hash index?
2. How does Linear Hashing work?
3. What are the pros and cons of different hash-based indexes?

Index Design Space

Organization Structure for k*

- Tree-based
  (+) Range, equality search
- Hash-based
  (+) Equality search
  - Static hashing
  - Extendible hashing
  - Linear hashing

Data Entry (k*) Contents

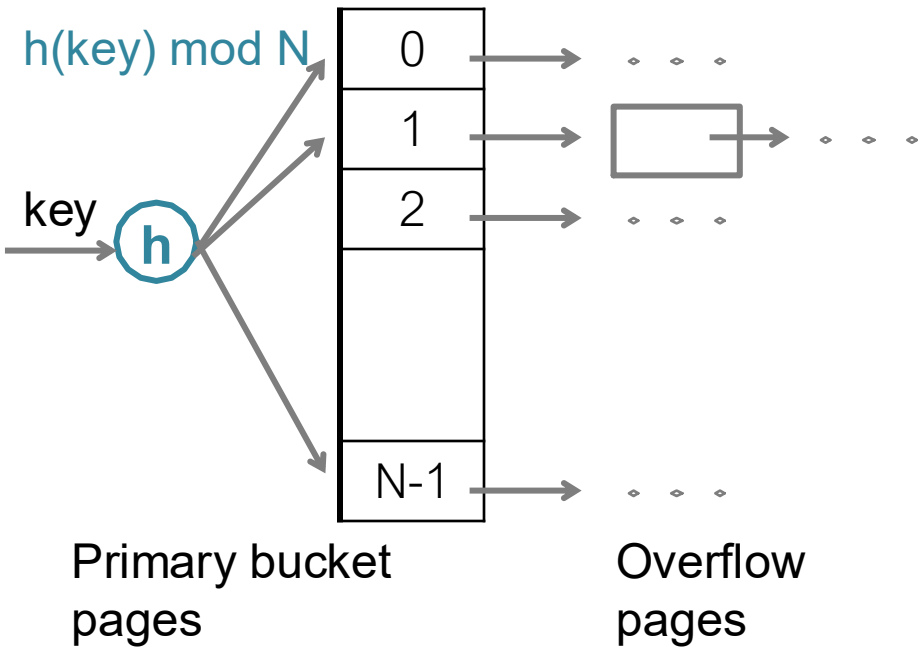
1. Actual Data record
   index = file
2. <k, rid>
   actual records in a diff file
3. <k, list of rids>

Hash Indexing in Practice

- Hash indexes support equality only, so most DBMS don't not offer user-facing hash index (CREATE HASH INDEX)
  - E.g., MySQL and SQLite mainly expose tree indexes
- Some DBs do provide user-visible hash indexes
  - Persistent: PostgreSQL (on-disk), MongoDB (hashed index)
  - In-memory: SQL Server supports hash index on memory-optimized tables
- Even without user facing hash index, DBMSs heavily use hashing for their internal operators:
  - hash join, hash aggregation, DISTINCT, and set operations.
  - Parallel DBMSs use shared hash tables and hash partitioning.

Static Hashing

- # primary bucket pages fixed, allocated sequentially, never de-allocated; overflow pages if needed.
- **h**(key) mod N = bucket to which data entry with key belongs. (N = # of buckets)



What are downsides of static hashing?

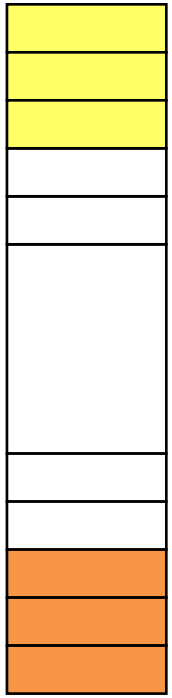
Static Hashing (Contd.)

- Buckets contain data entries
- Number of buckets (N) is fixed ahead of time
- Static structure can be problematic
  - Consider many insertions
  - Long overflow chains can develop (and degrade performance!)
- Might consider periodically doubling N and "rehashing" file
  - Entire file has to be read and written
  - Index unavailable while rehashing
  - Dynamic hashing fixes this problem: Linear and Extendible Hashing

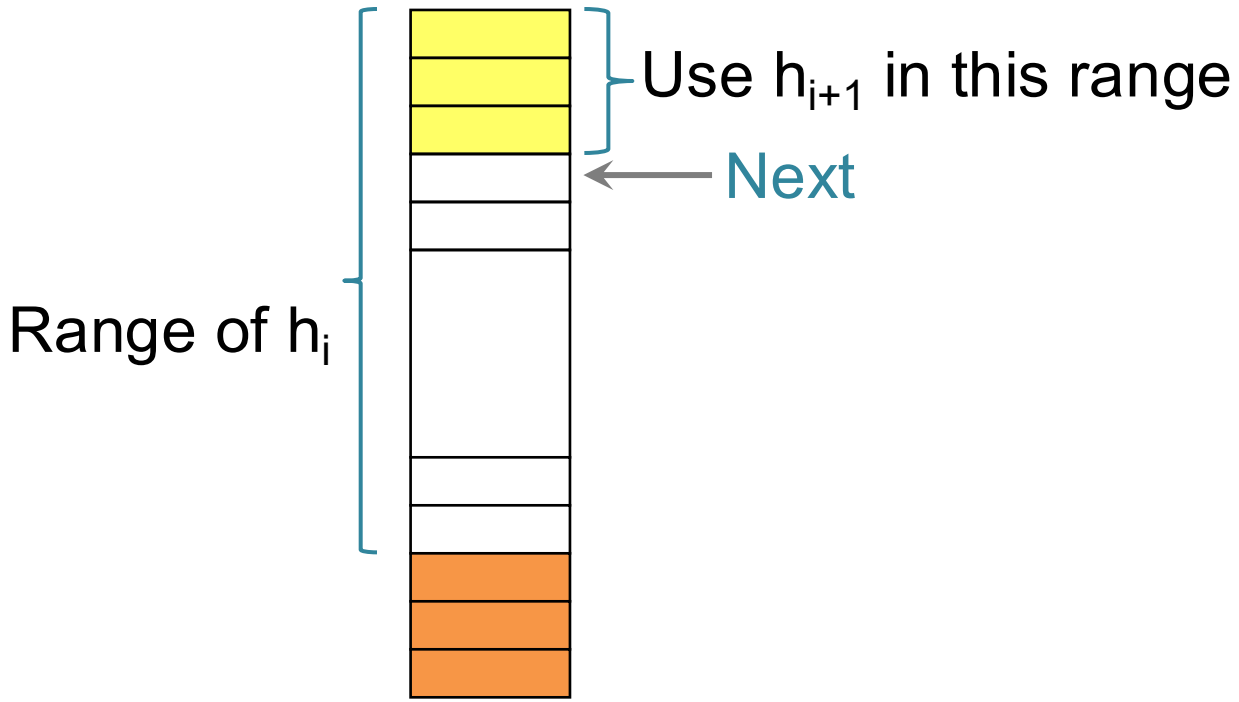
Why not start w/ a very large N?

Linear Hashing

- Works in rounds
- Grows gradually
  - More gradually than other dynamic hashing such as extendible hashing



Buckets During Round *i*



Split image buckets (created by splitting other buckets in this round)

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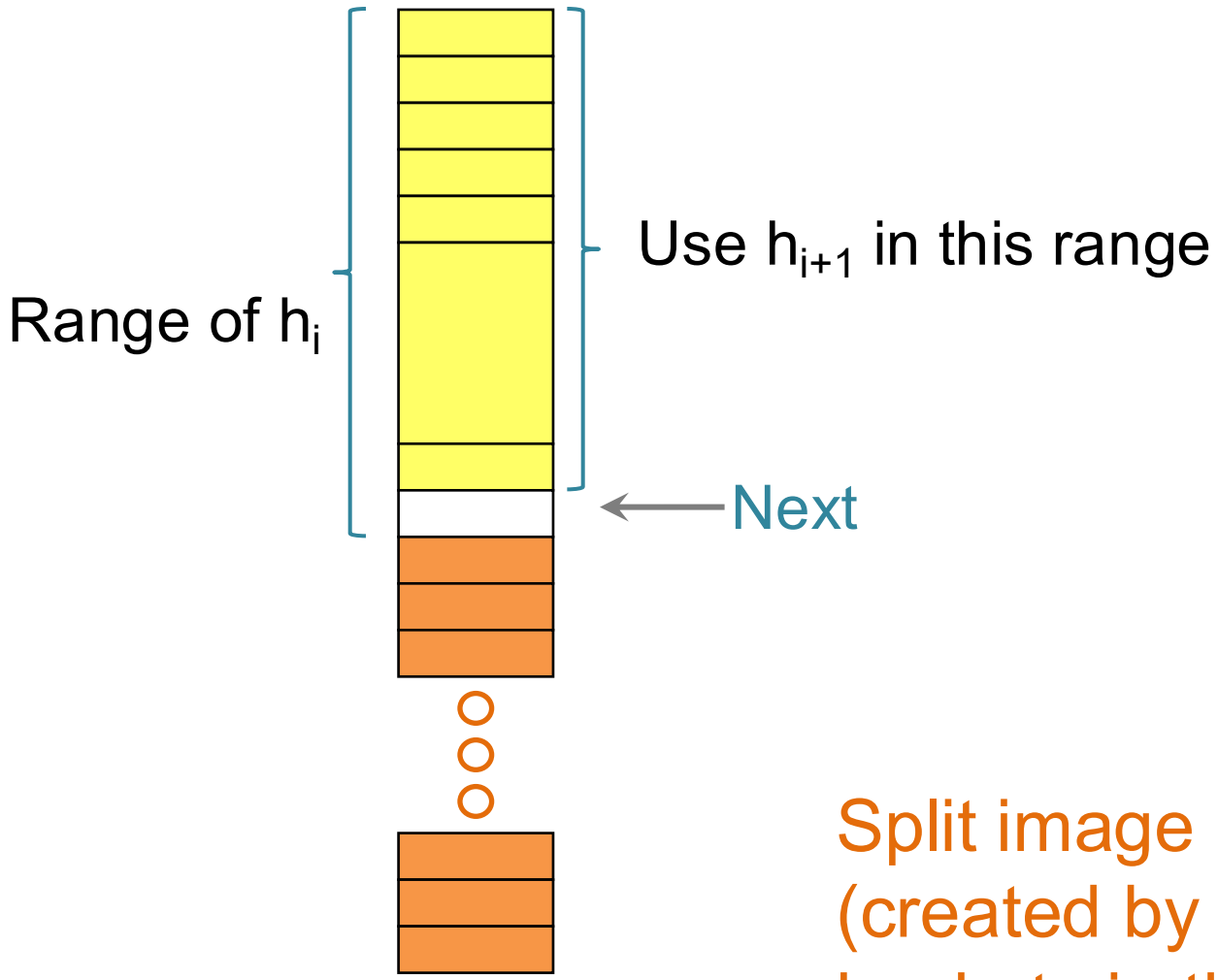
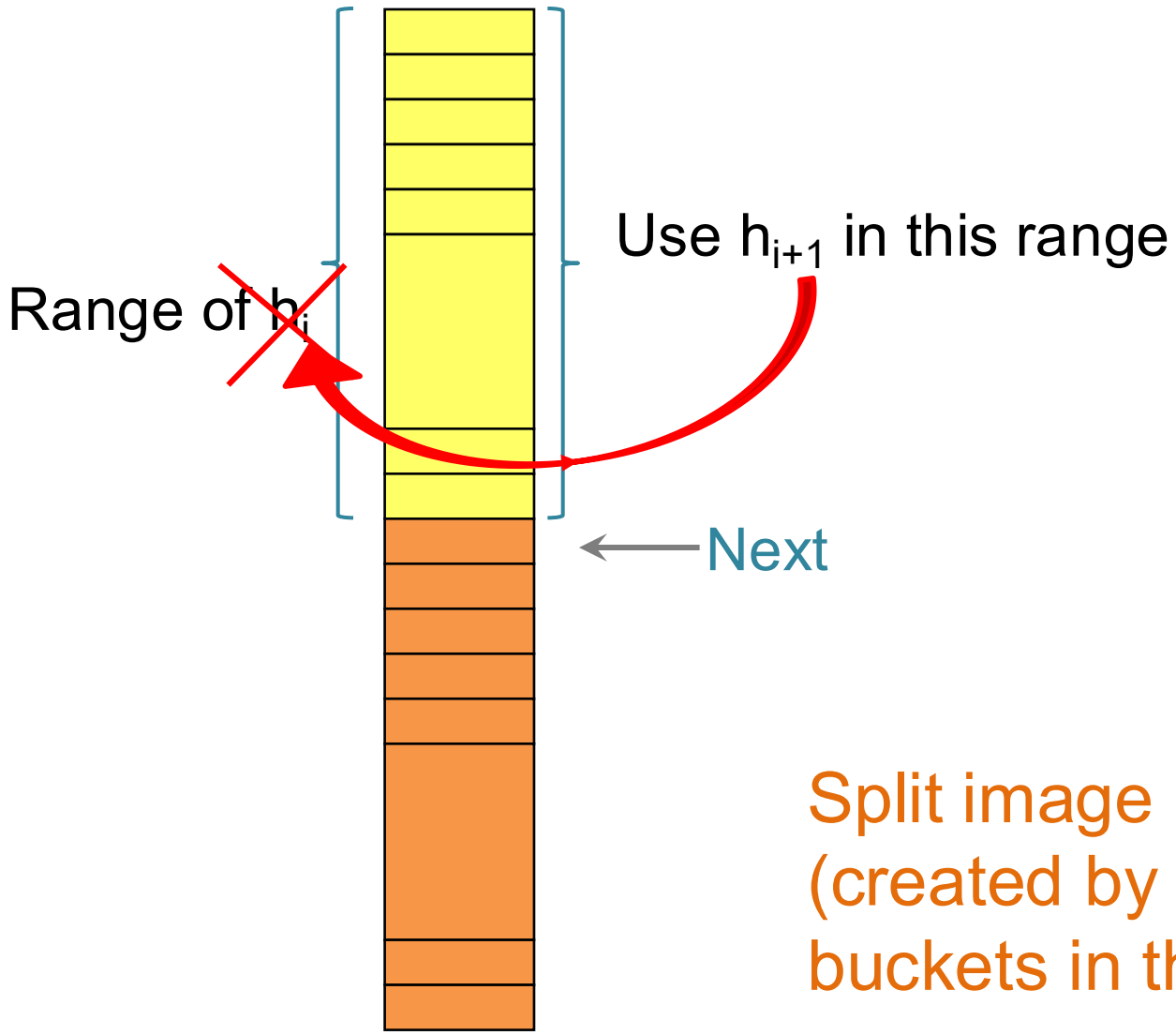


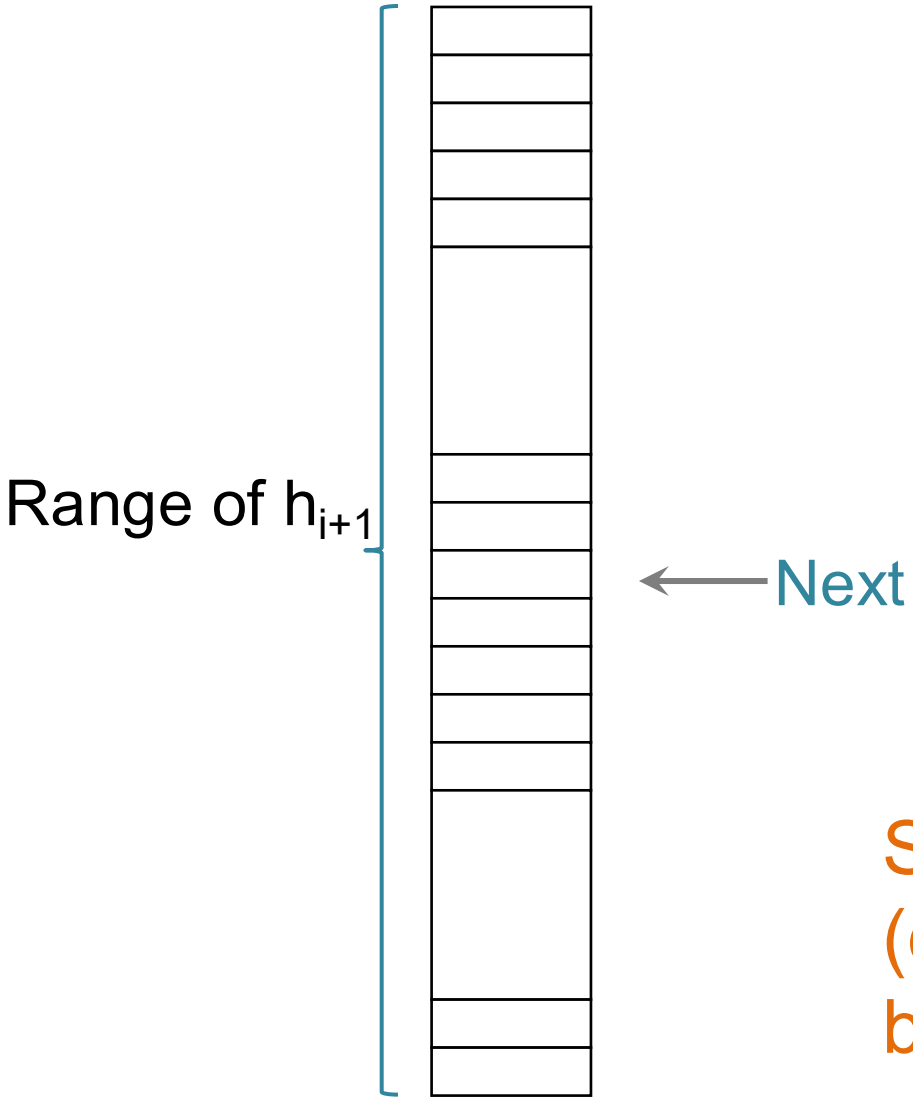
Split image buckets (created by splitting other buckets in this round)

End of Round *i* = Start of Round *i+1*



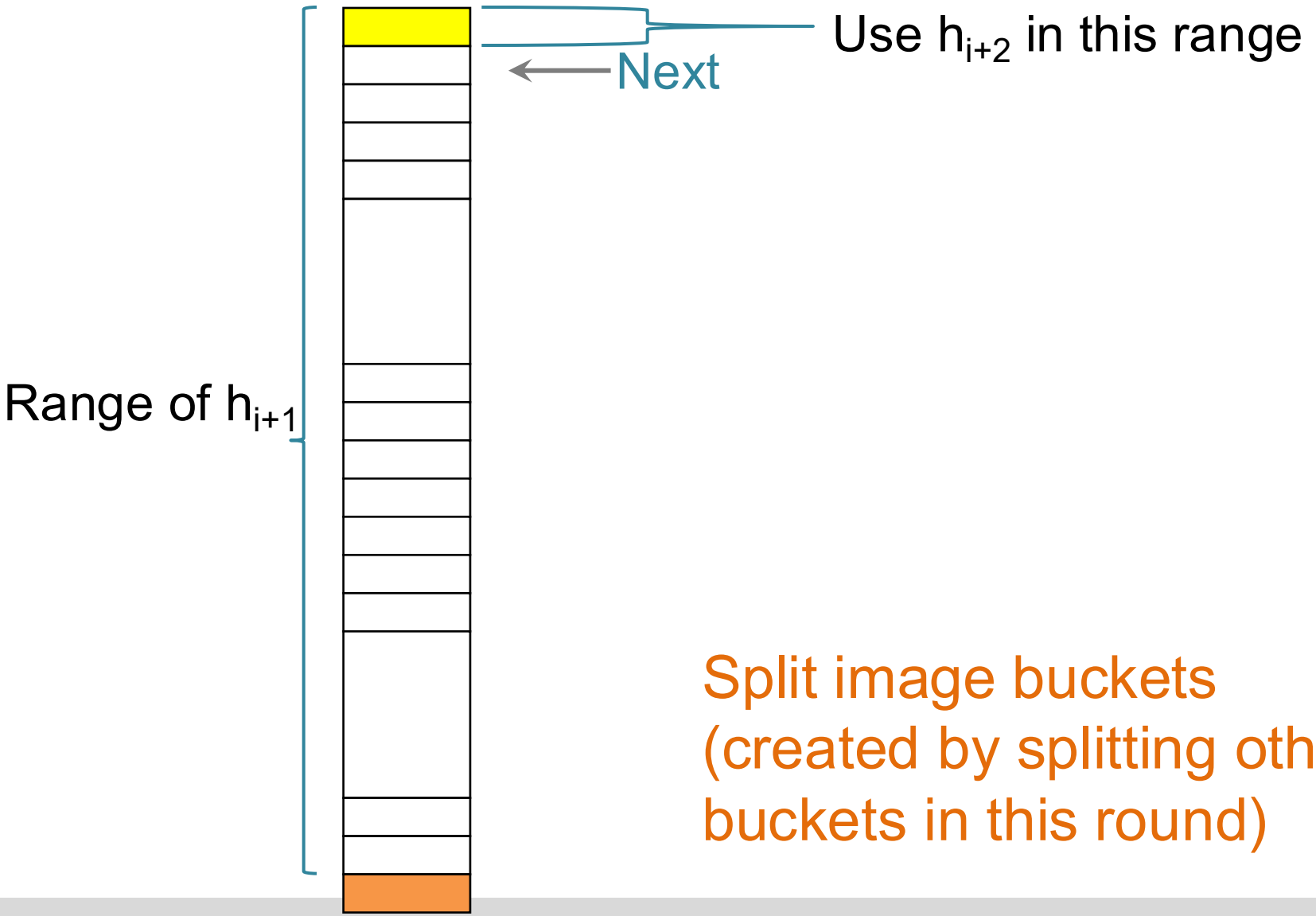
Split image buckets (created by splitting other buckets in this round)

End of Round *i* = Start of Round *i+1*



Split image buckets (created by splitting other buckets in this round)

Round *i+1* after the First Split



Split image buckets (created by splitting other buckets in this round)

Linear Hashing: Main Idea

- Another dynamic hashing scheme
- Eliminates long overflow chains without using a directory
- **Idea:** Use a family of hash functions: $\mathbf{h_0}$, $\mathbf{h_1}$, $\mathbf{h_2}$, ...
  - h_{i+1} doubles the range of h_i (similar to directory doubling)
  - Hash family typically obtained by choosing hash function *h*( ) and initial number of buckets *N*
  - Define $h_i(value) = h(value) \; mod \; (2^i N)$
  - If N is a power of 2, say $\mathbf{N=2^{d_0}}$, apply hash function *h*( ), and look at last $\boldsymbol{d_i}$ bits where $\mathbf{d_i = d_0 + i}$

Linear Hashing: Details

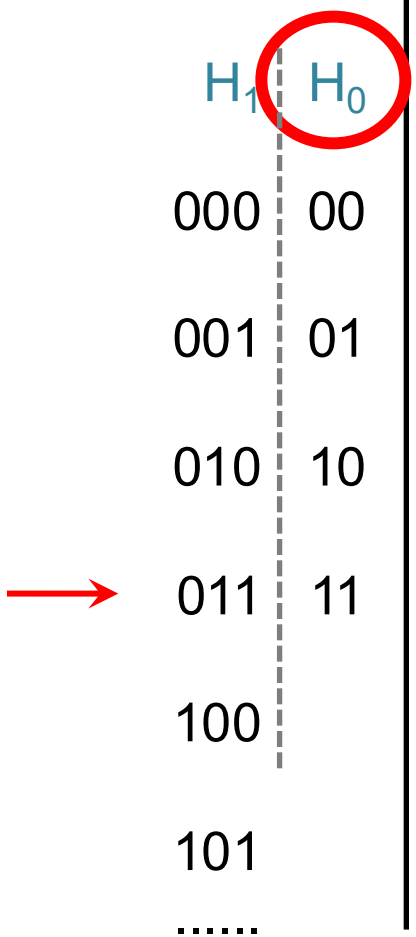
- Splitting proceeds in rounds
  - During round *Level*, only $\mathbf{h}_{Level}$ and $\mathbf{h}_{Level+1}$ are in use
- Variables
  - Level: Initialized to 0
  - Next: Pointer to the bucket being split
- At the beginning of round # Level,
  - # buckets in the file = $N_{Level} = N * 2^{Level}$

  where N is initial number of buckets

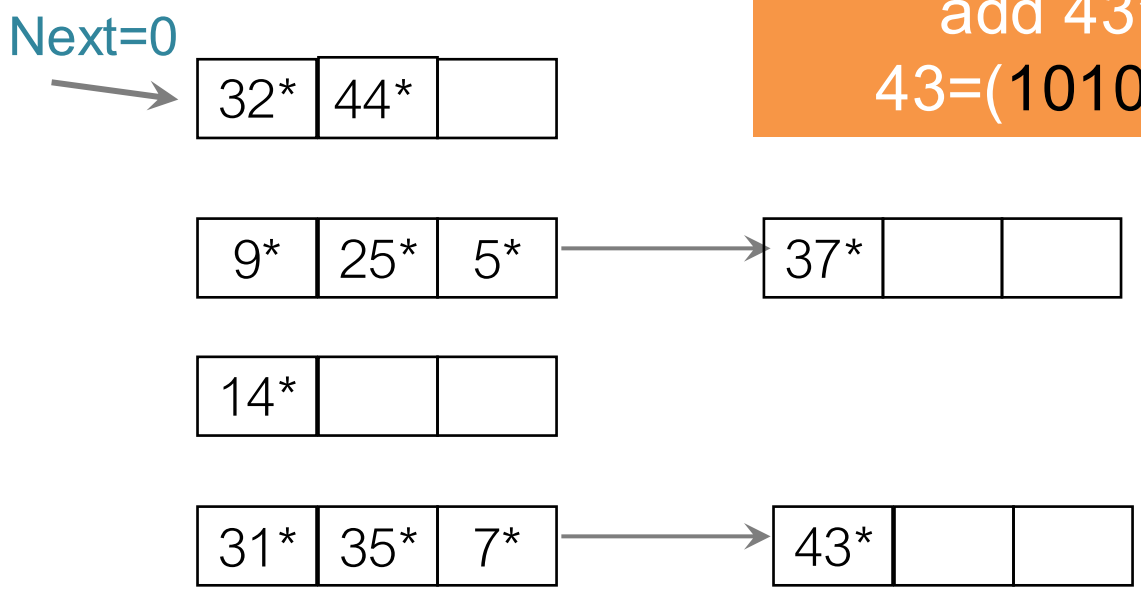
Linear Hashing Example

Level 0, N = 4

What happens if we add 43*? $43=(101011)_2$



This is not actually stored

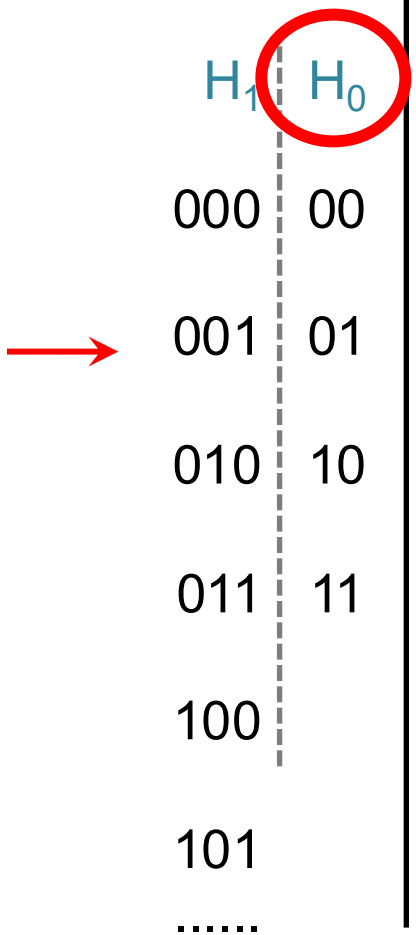


Note: Here, the split bucket is different than the one triggering the split

Linear Hashing Example

Level 0, N = 4

What happens if we now insert 17*? $17=(10001)_2$



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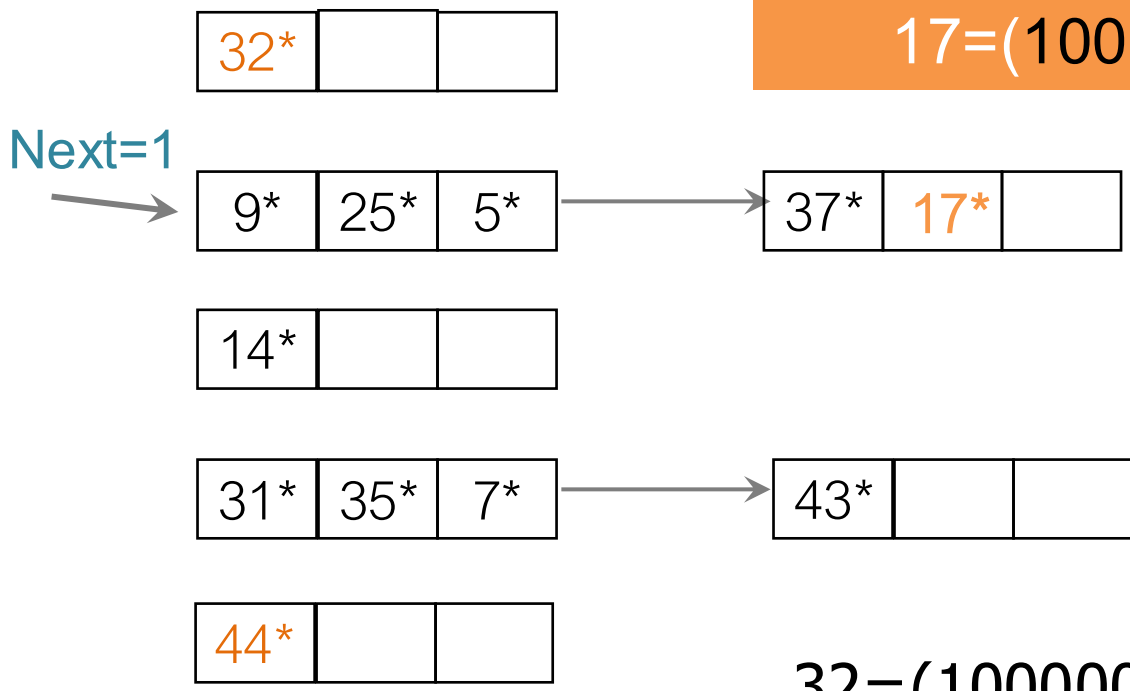
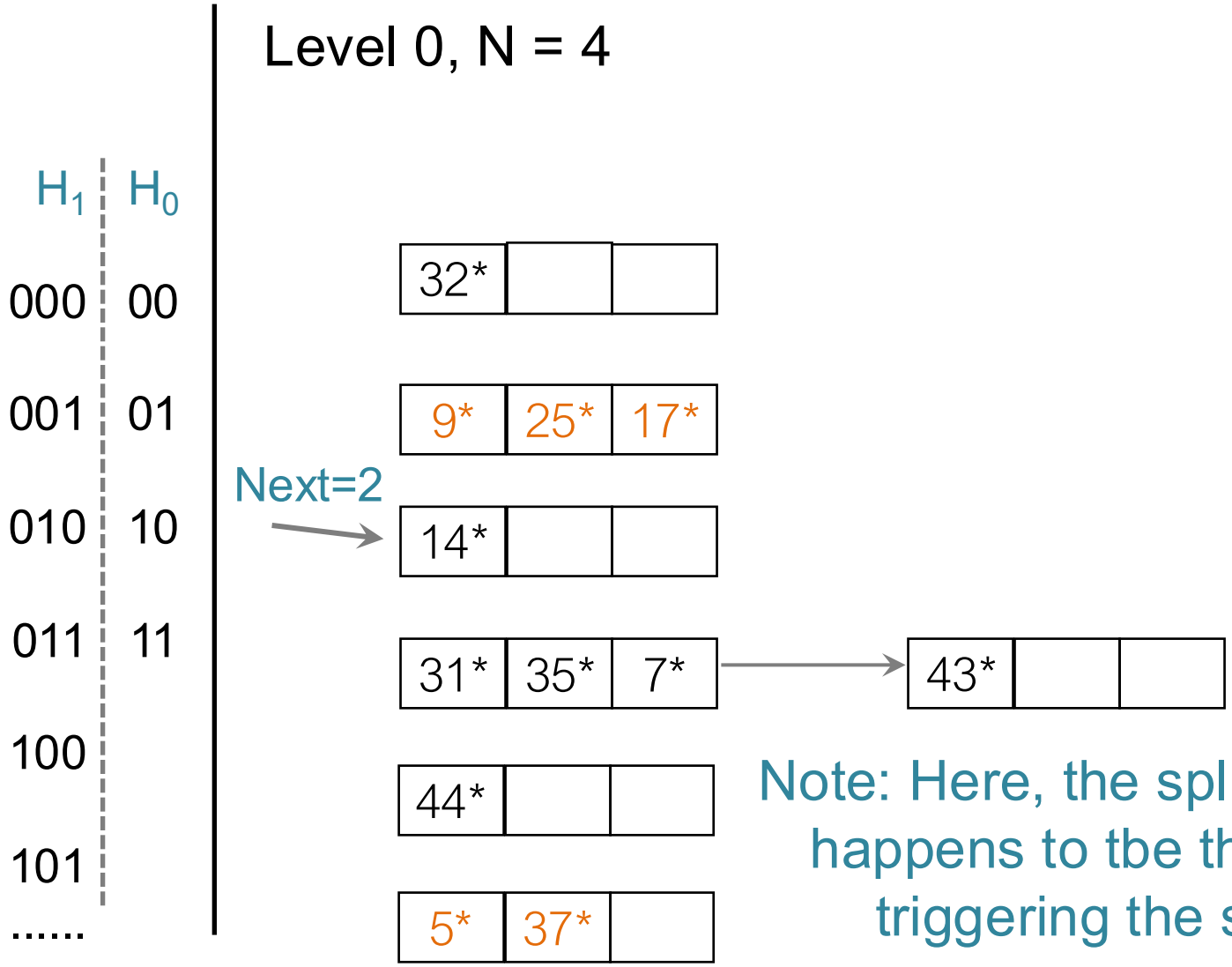


$32=(100000)_2$
$44=(101100)_2$

Linear Hashing Example



Note: Here, the split bucket happens to tbe the one triggering the split

Linear Hashing Algorithm

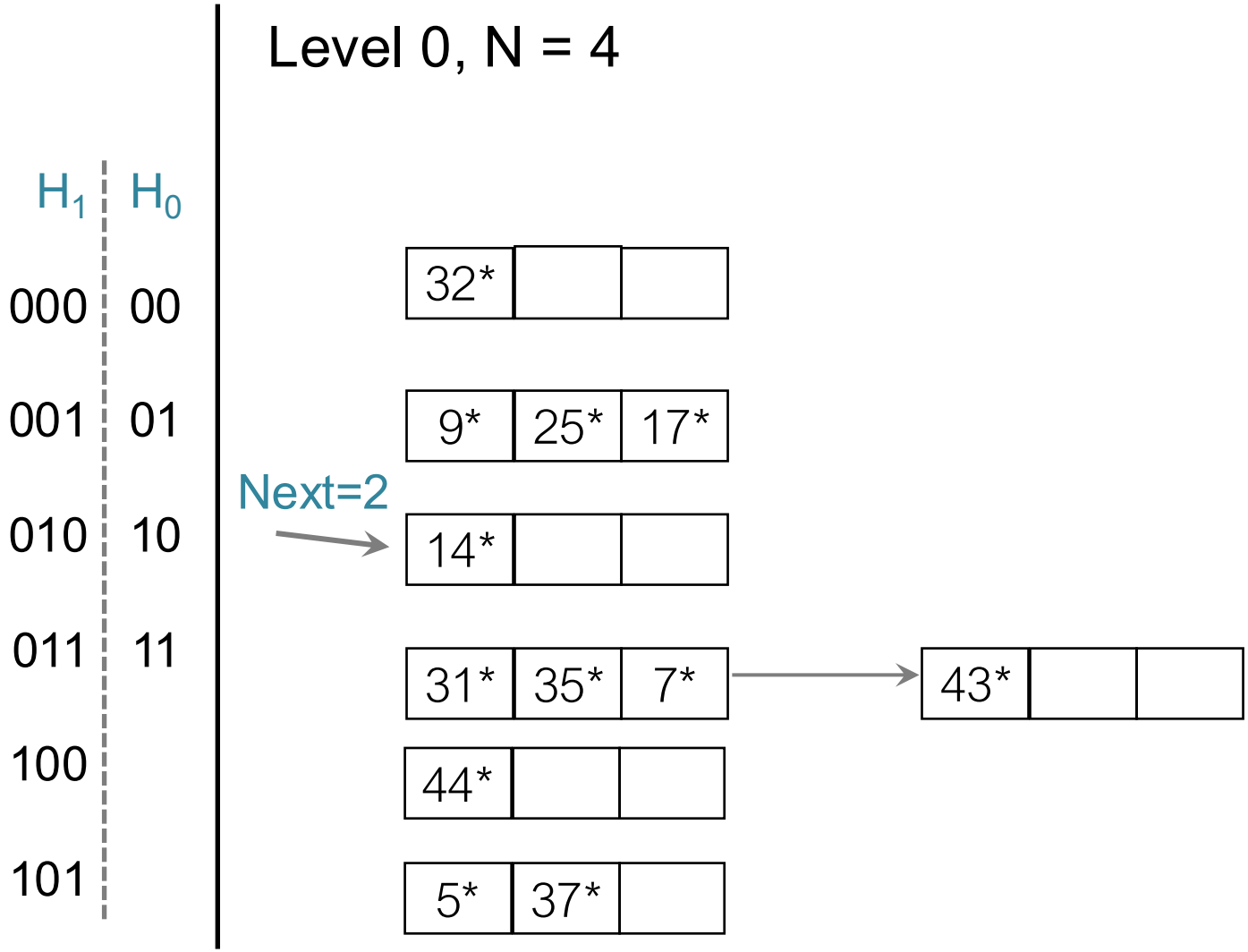
Insert:

- Find bucket by applying $\mathbf{h}_{Level}$
  - Apply $\mathbf{h}_{Level+1}$ if the value returned by $\mathbf{h}_{Level}$ < Next
- If bucket being inserted into is full:
  - Add overflow page and insert data entry
    - If bucket is Next, split first and then see if you still need overflow
  - Split *Next* bucket and increment Next
    - If Next = N_{Level} – 1 and a split is triggered
      1. We split the bucket # N_{Level}-1 (i.e. the last bucket before the split image buckets)
      2. Next = 0 (reset the pointer)
      3. Level = Level + 1 (the size has doubled at this point)

Linear Hashing (Contd.)

- Can choose any criterion to trigger split
  - e.g., split on adding to an overflow page (as in example)
  - e.g., space utilization on the page > 90%
- Since buckets are split round-robin, long overflow chains don't develop!
- Deletes: see textbook

Quiz: Insert 13=$(1101)_2$ & 29=$(11101)_2$



Summary

- Static Hashing can lead to long overflow chains
- Extendible Hashing
  - Directory to keep track of buckets, doubles periodically
  - Always splits the “right” bucket
- Due to their inability to support range queries, many DBMSs don’t support user-facing hash indexing. Nonetheless, almost all DBMSs use them internally

Optional Exercises

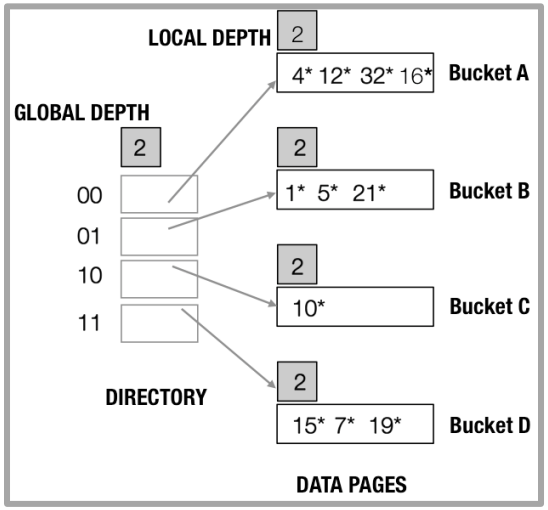
11.1, 11.3, 11.7, 11.9

Appendix: Optional Slides on Extendible Hashing

Extendible Hashing

- Idea: Use a directory of pointers to buckets

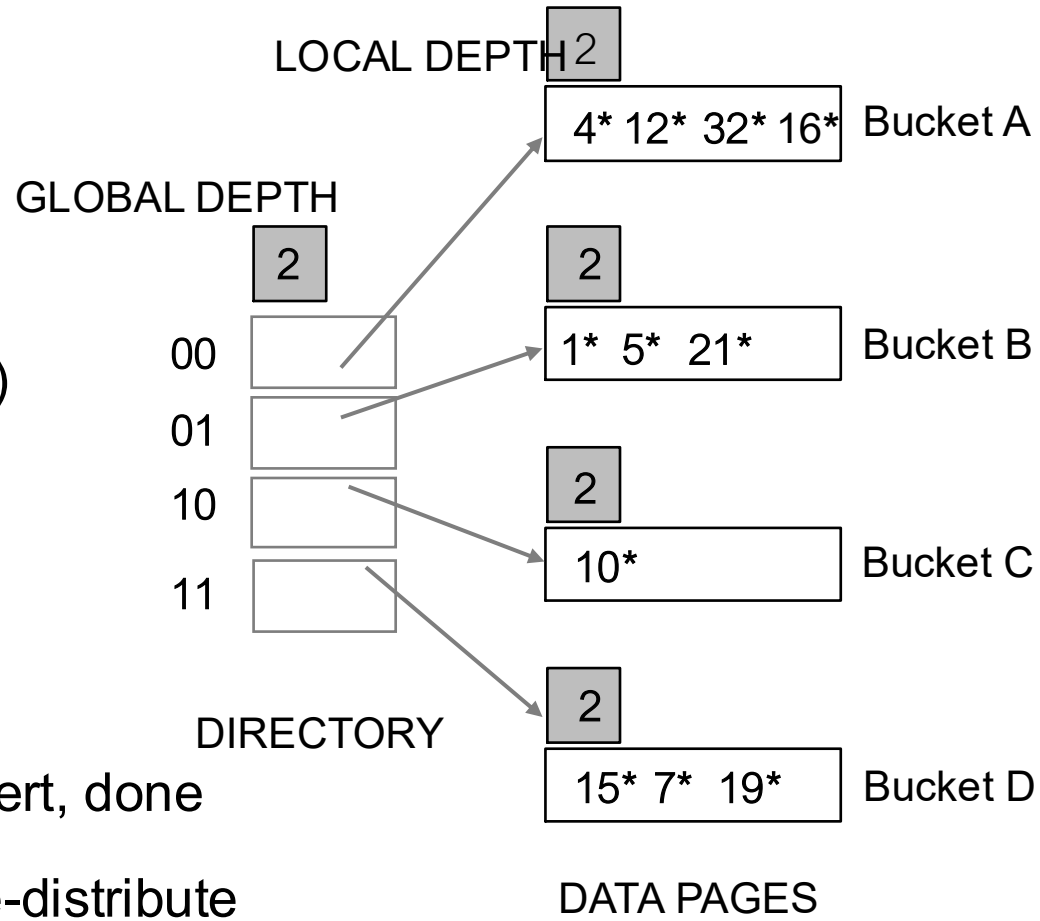
1. On overflow, split the bucket
2. If new bucket has no pointers in directory, double the directory (not # number of buckets)



- Why does this help?
  - Directory much smaller than the file
  - Only one page of data entries is split at a time
  - (Usually*) No overflow pages

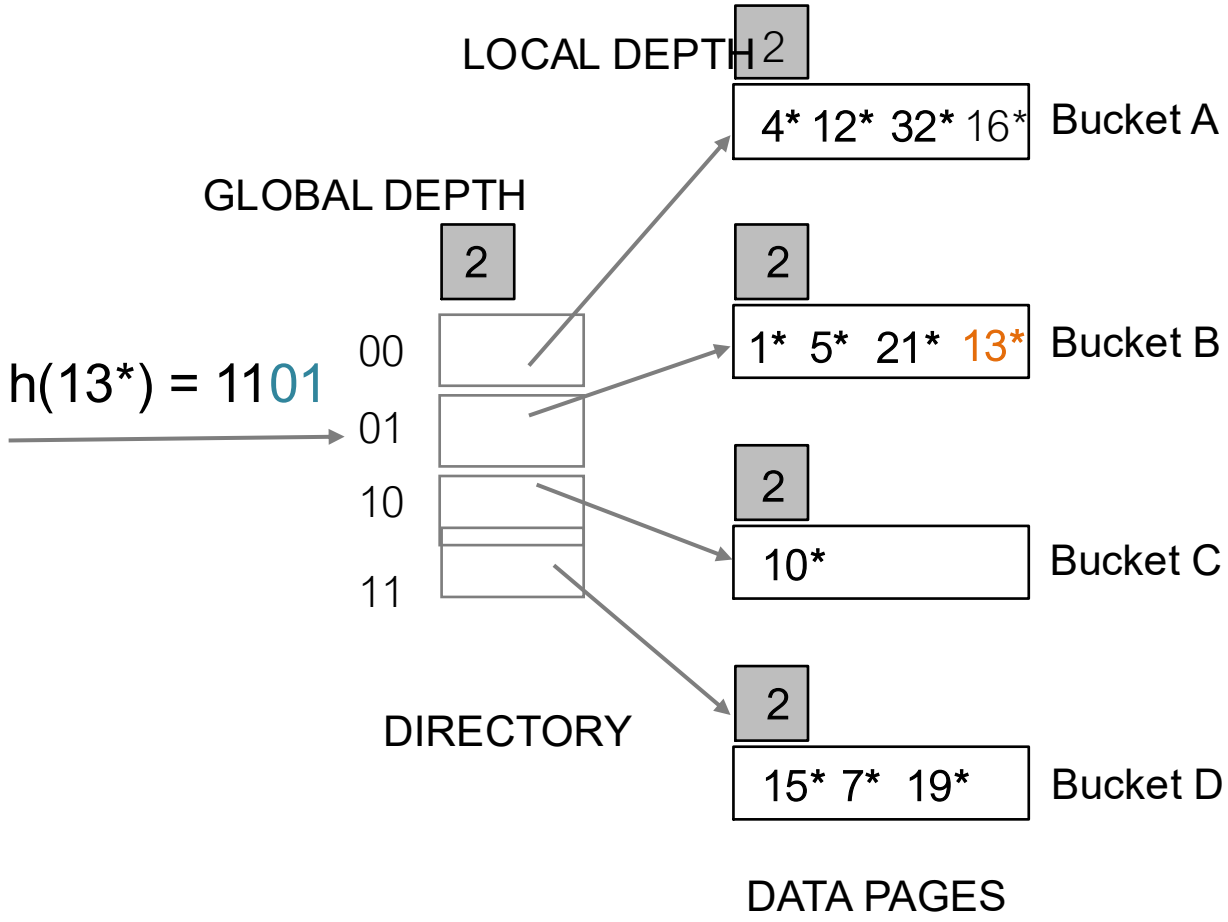
Extendible Hashing: Algorithm

- Directory = an array
- Search for k:
  - Apply hash function h(k)
  - Take last global depth # bits of **h**(*k*)
- Insert:
  - If bucket has space, insert, done
  - If bucket if full, split it, re-distribute
  - If necessary, double the directory

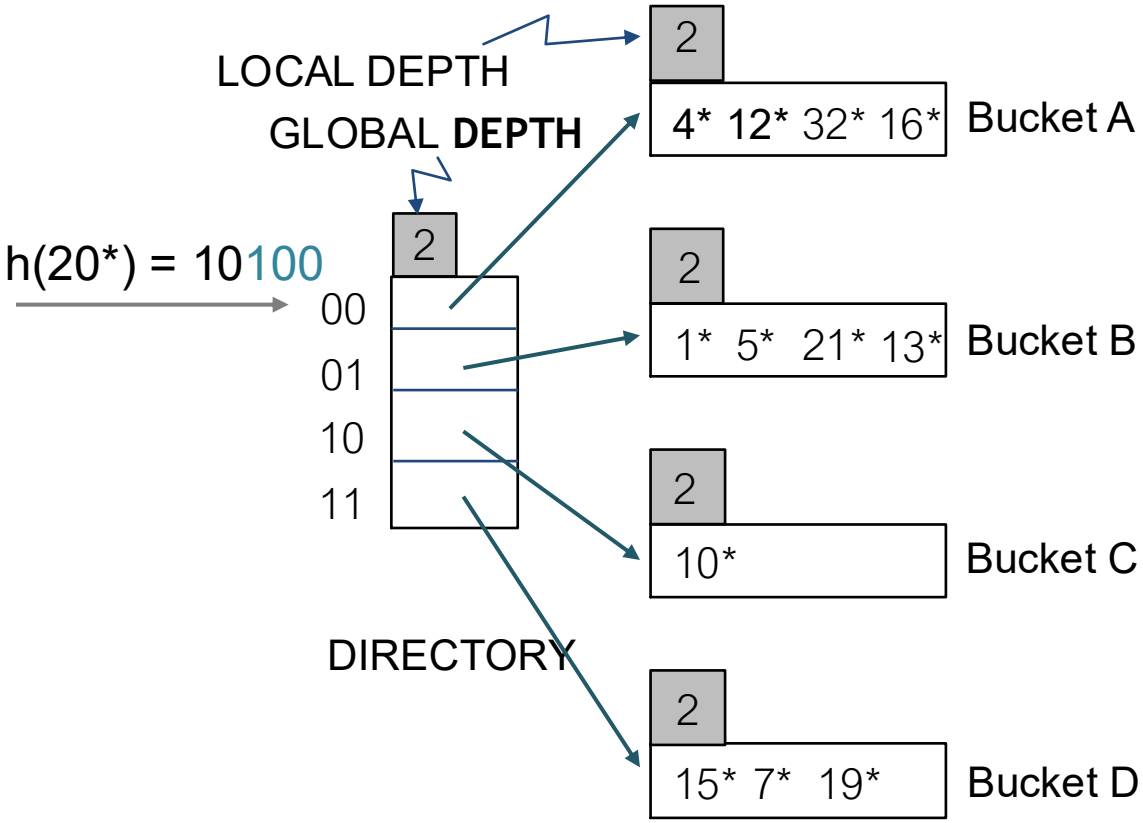


Example: Insertion into a free bucket

- Insert 13*
- Suppose
  h(13*) = 1101



Example: Insertion into a full bucket
Insert 20 = $(10100)_2$



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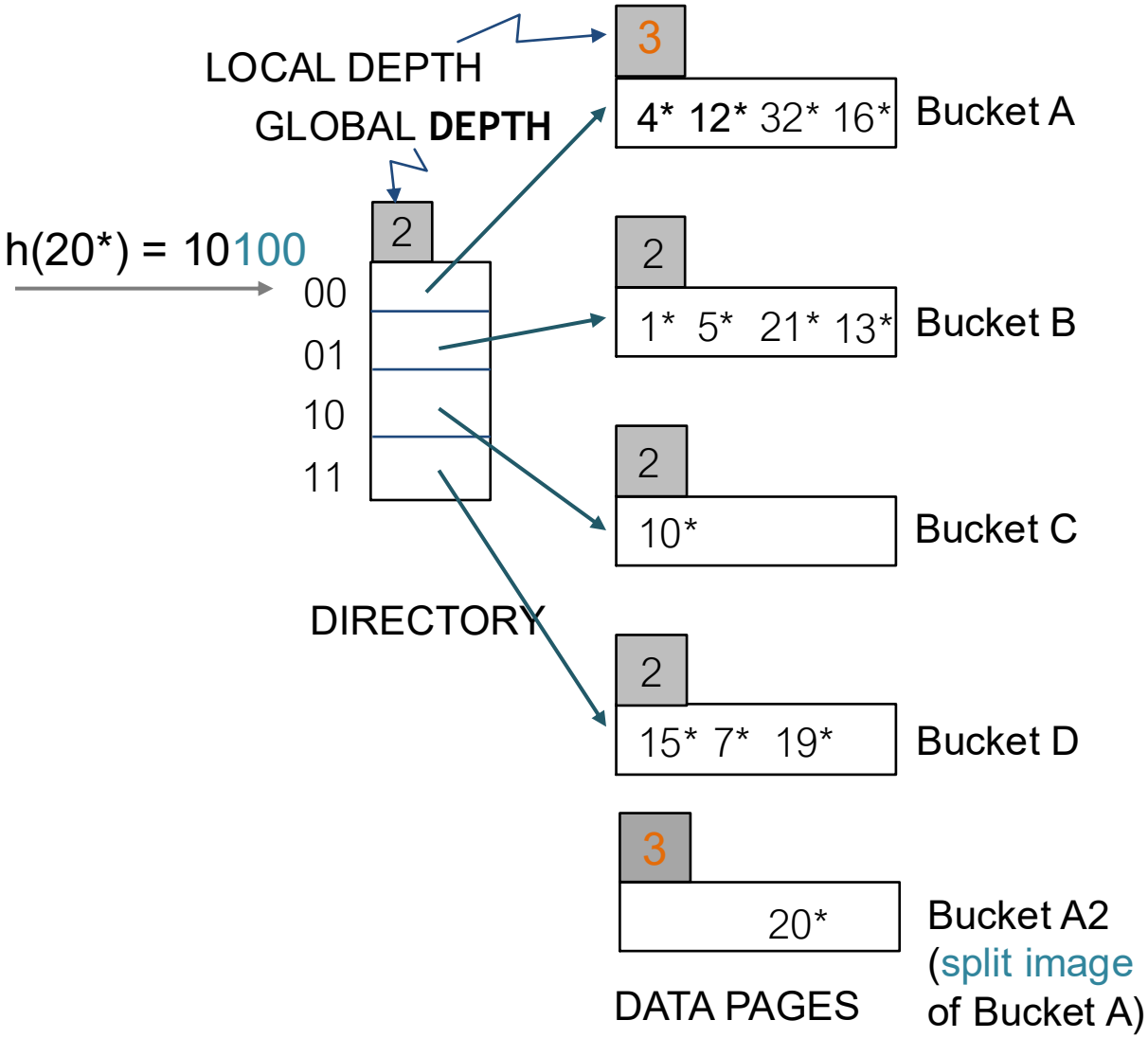

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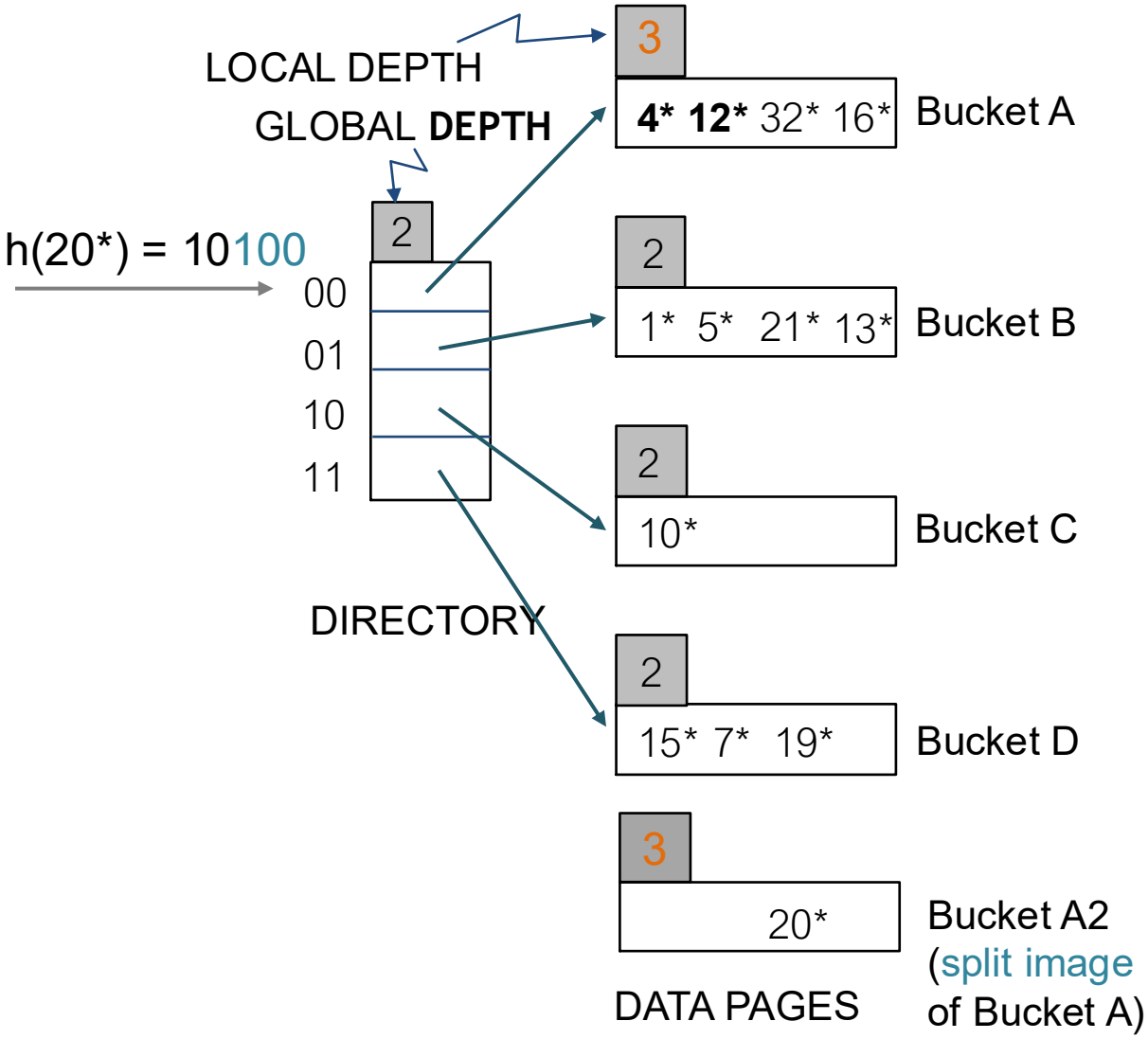

$32=(100000)_2$
$16=(10000)_2$

$4=(100)_2$
$12=(1100)_2$
$20=(10100)_2$

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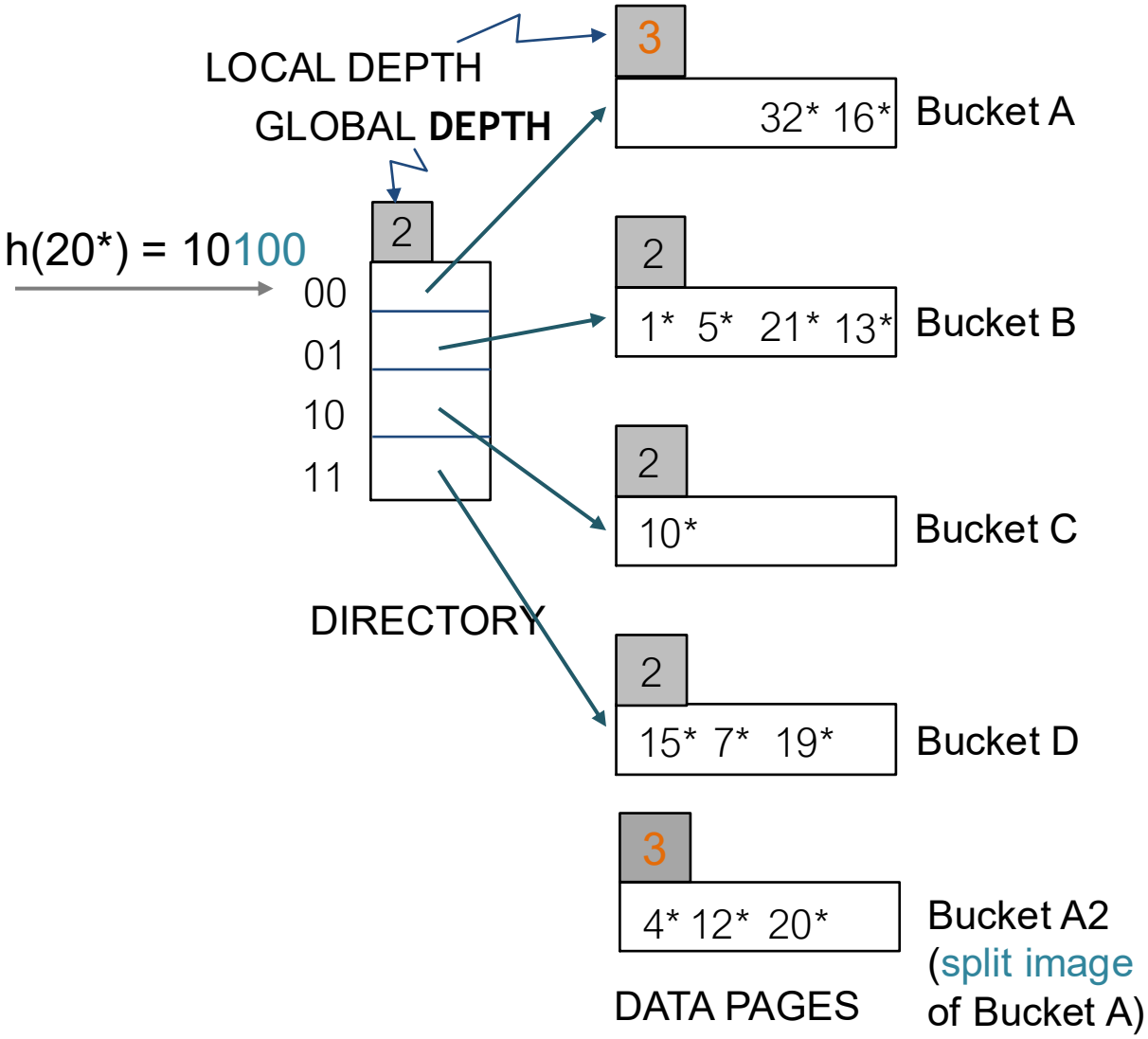

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Example: Insertion into a full bucket
Insert 20 = $(10100)_2$

LOCAL DEPTH

GLOBAL **DEPTH**

$h(20^*) = 10100$

DIRECTORY

3	32* 16*	Bucket A
2	1* 5* 21* 13*	Bucket B
2	10*	Bucket C
2	15* 7* 19*	Bucket D
3	4* 12* 20*	Bucket A2 (split image of Bucket A)

DATA PAGES

$32=(100000)_2$
$16=(10000)_2$

$4=(100)_2$
$12=(1100)_2$
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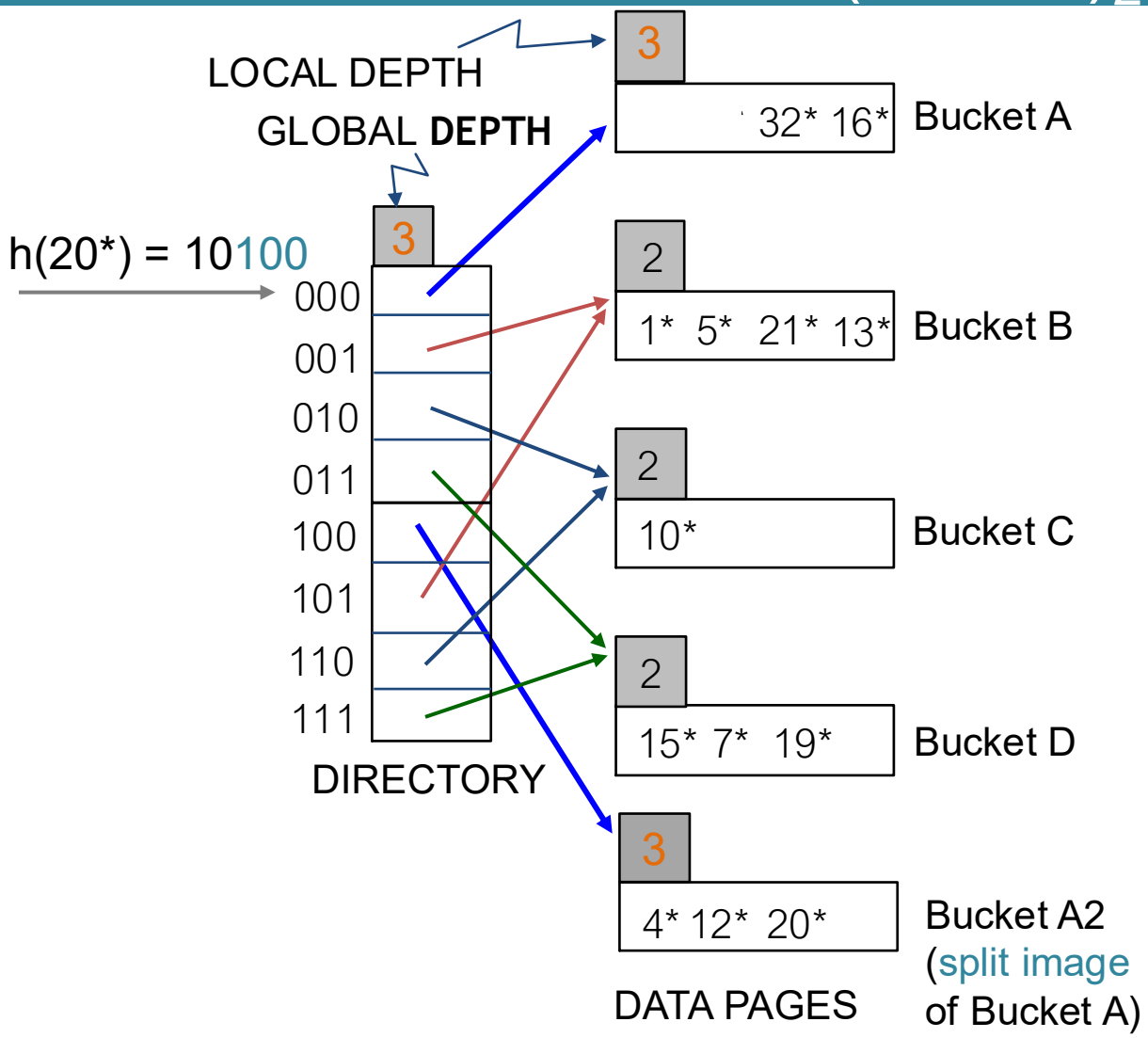
Example: Insertion into a full bucket
Insert 20 = $(10100)_2$



$32=(100000)_2$
$16=(10000)_2$

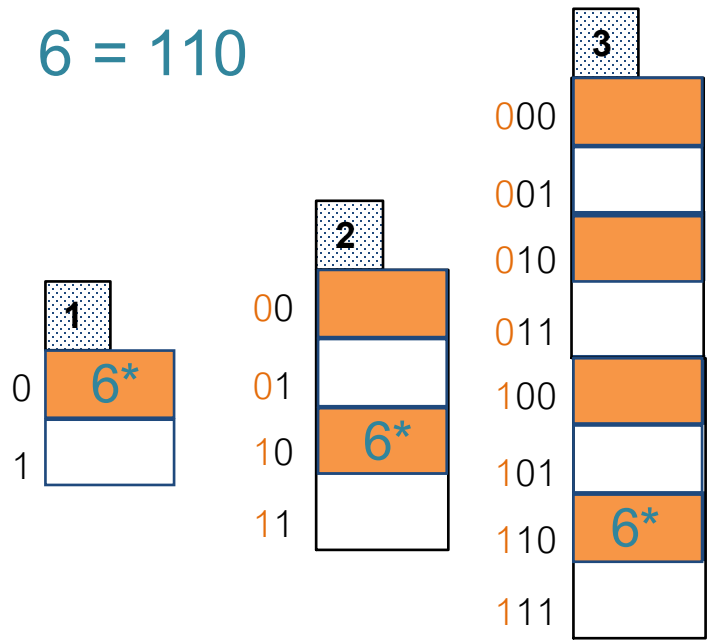
- Initially, all LDs = GD
- Splitting a bucket increments its LD
- GD should always be >= all LDs
- Hence, splitting a bucket whose LD = GD, doubles the directory
- Directory doubled by *copying it over* and fixing pointer to split image page

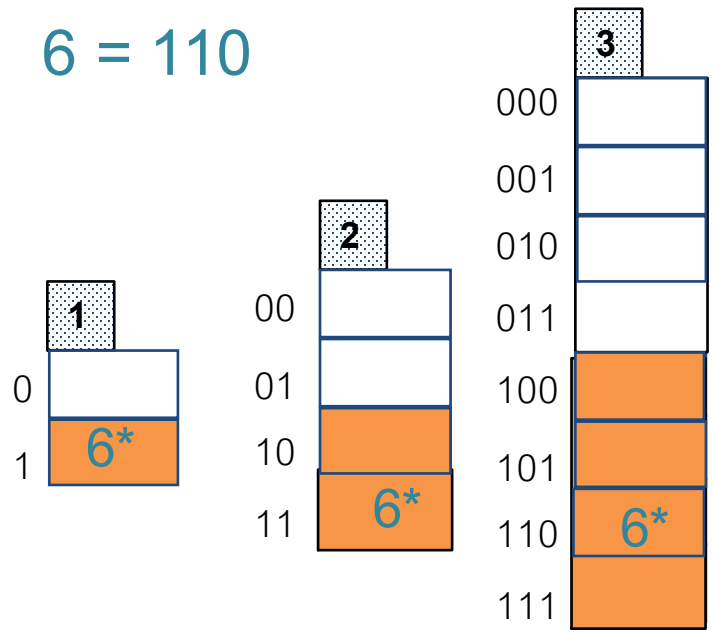
$4=(100)_2$
$12=(1100)_2$
$20=(10100)_2$

Directory Doubling

Why use least significant bits in directory?

Allows for doubling via copying!



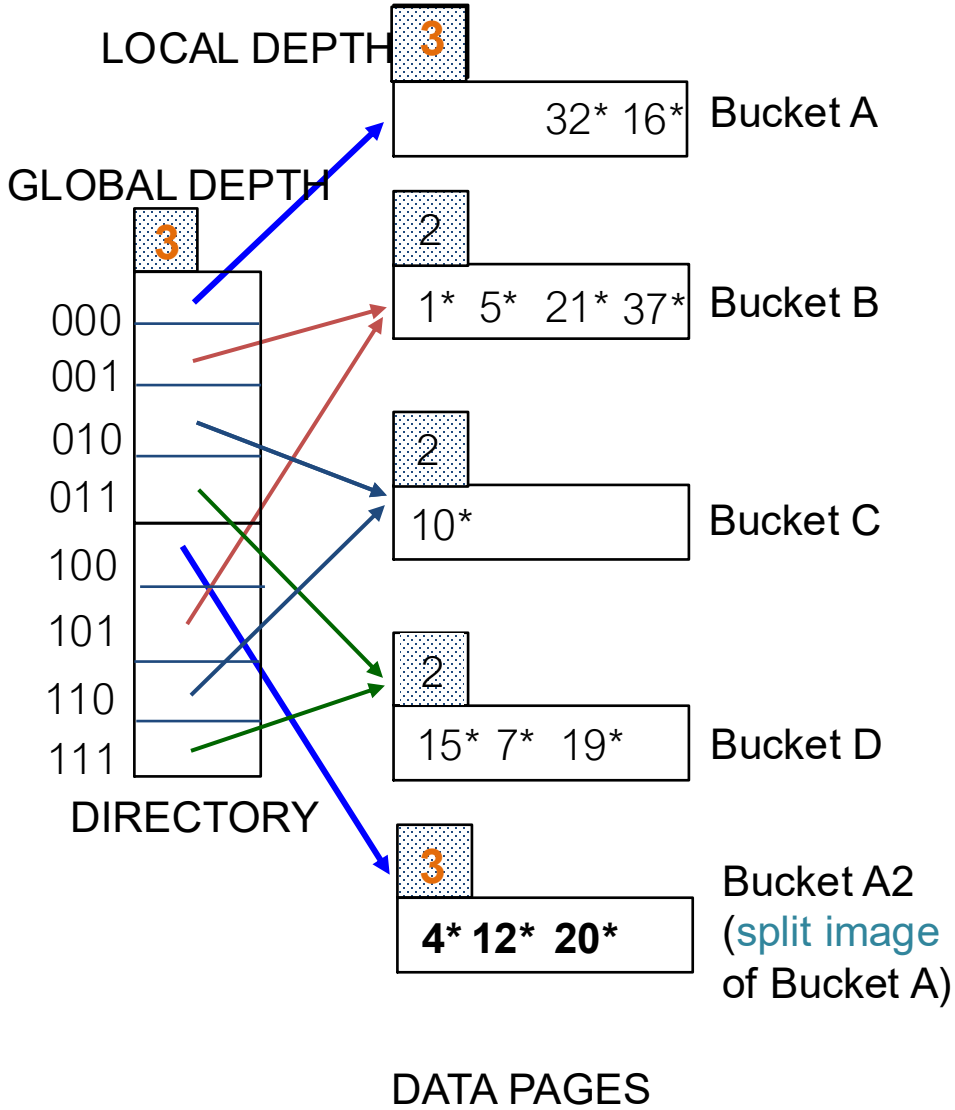


Least Significant vs. Most Significant

Comments on Extendible Hashing

- How many disk accesses for equality search?
  - One if directory fits in memory, else two
- Directory grows in spurts, and, if the distribution *of hash values* is skewed, directory can grow large
- Do we ever need overflow pages?
  - Multiple entries with same hash value cause problems! => We need overflow pages
- Delete: Reverse of inserts – see textbook

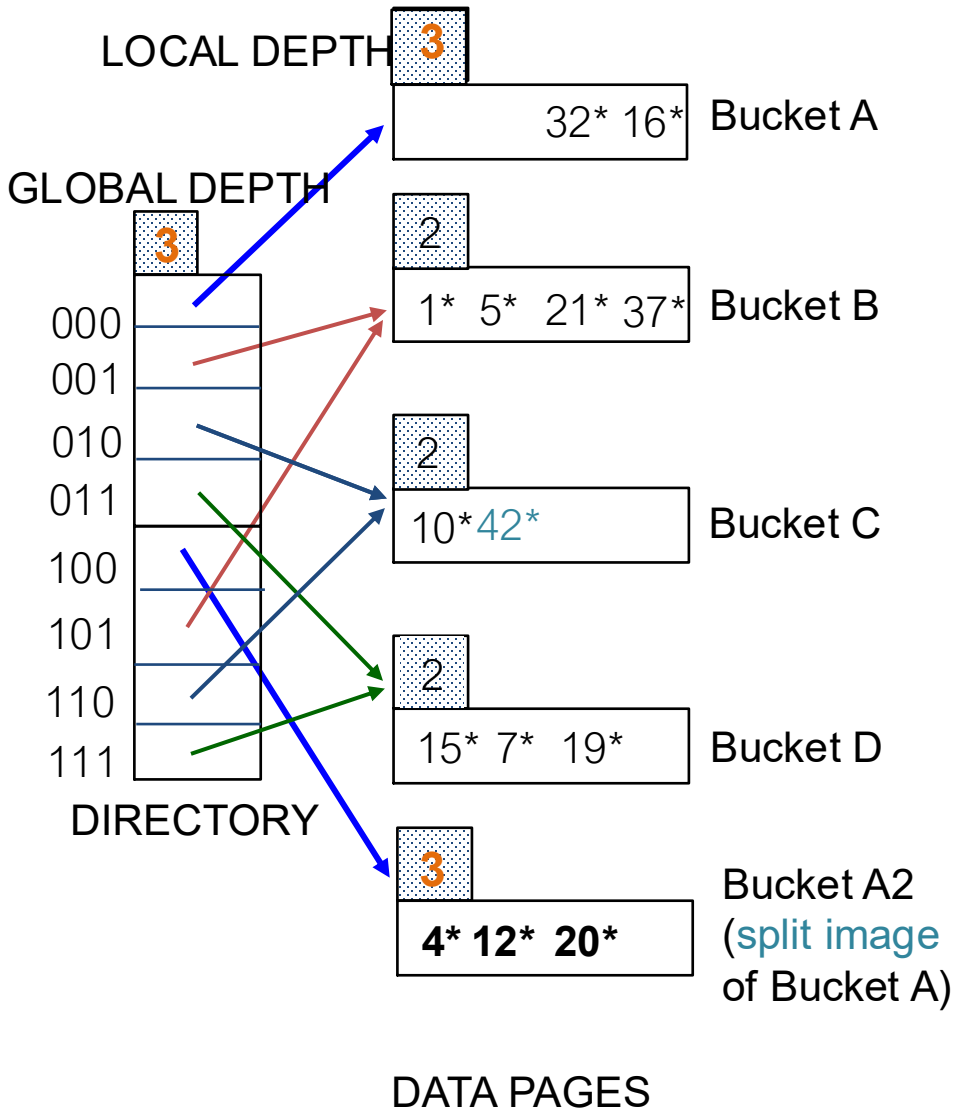
Extendible Hashing Quiz: Insert 42, 9



$42=(101010)_2$

$9=(1001)_2$

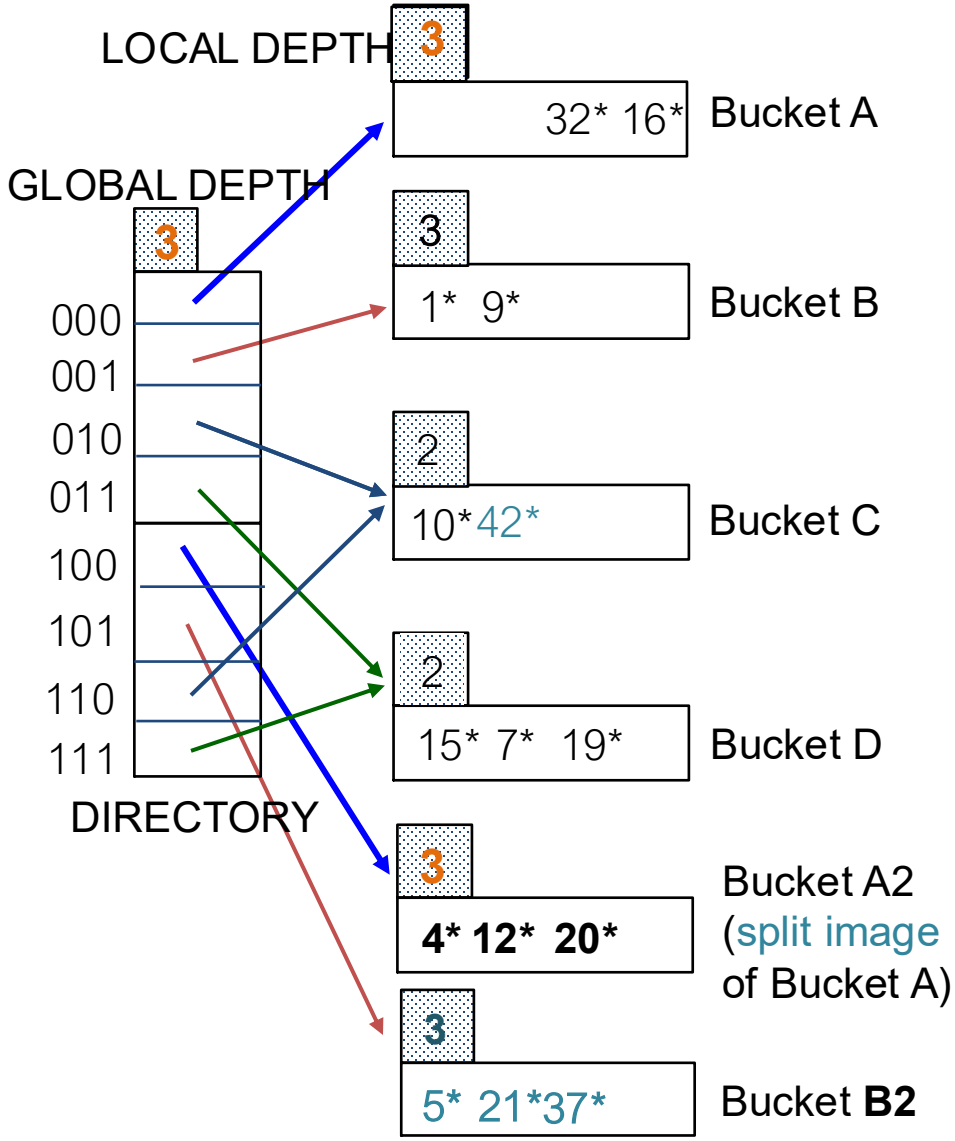
Answer: Insert 42



$42=(101010)_2$

$9=(1001)_2$

Answer: Insert 9



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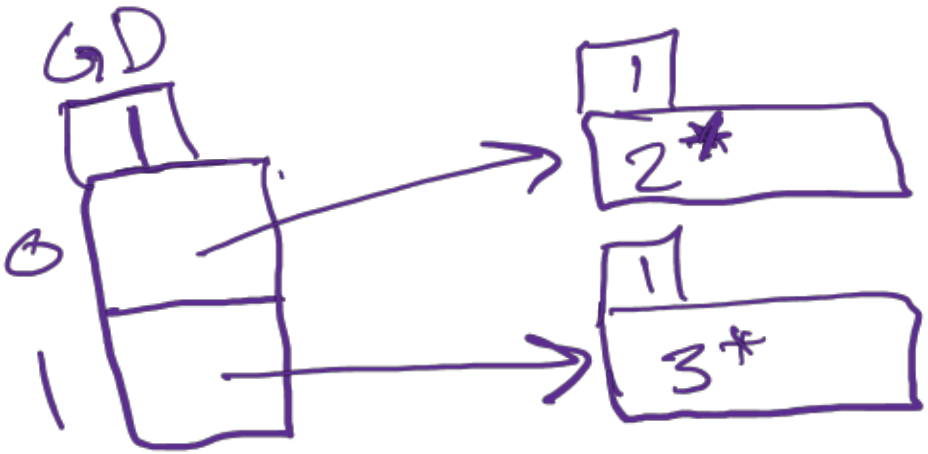
Quiz: insert 11 and 27



$3 = (11)_2$

$11 = (1011)_2$

$27 = (11011)_2$

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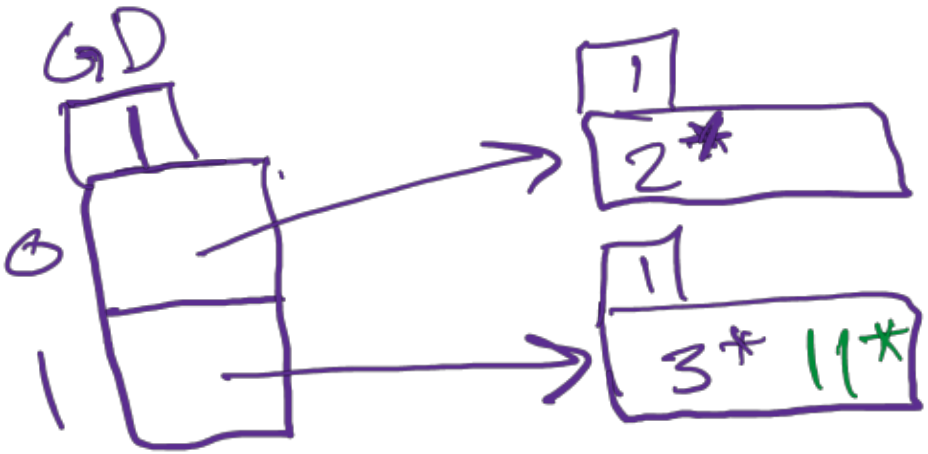

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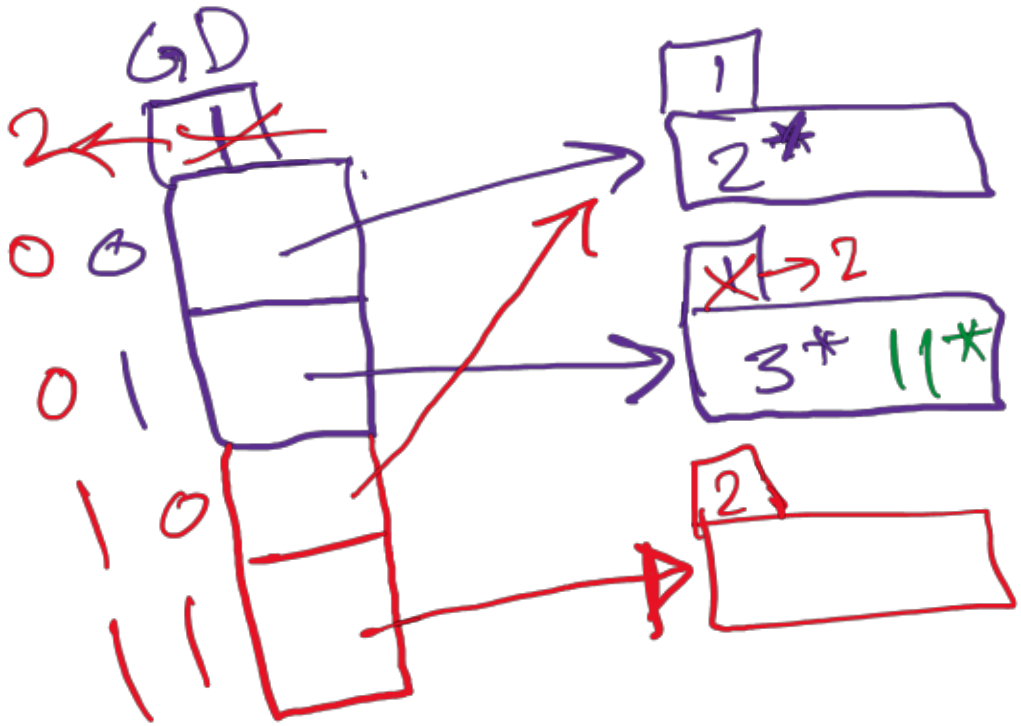

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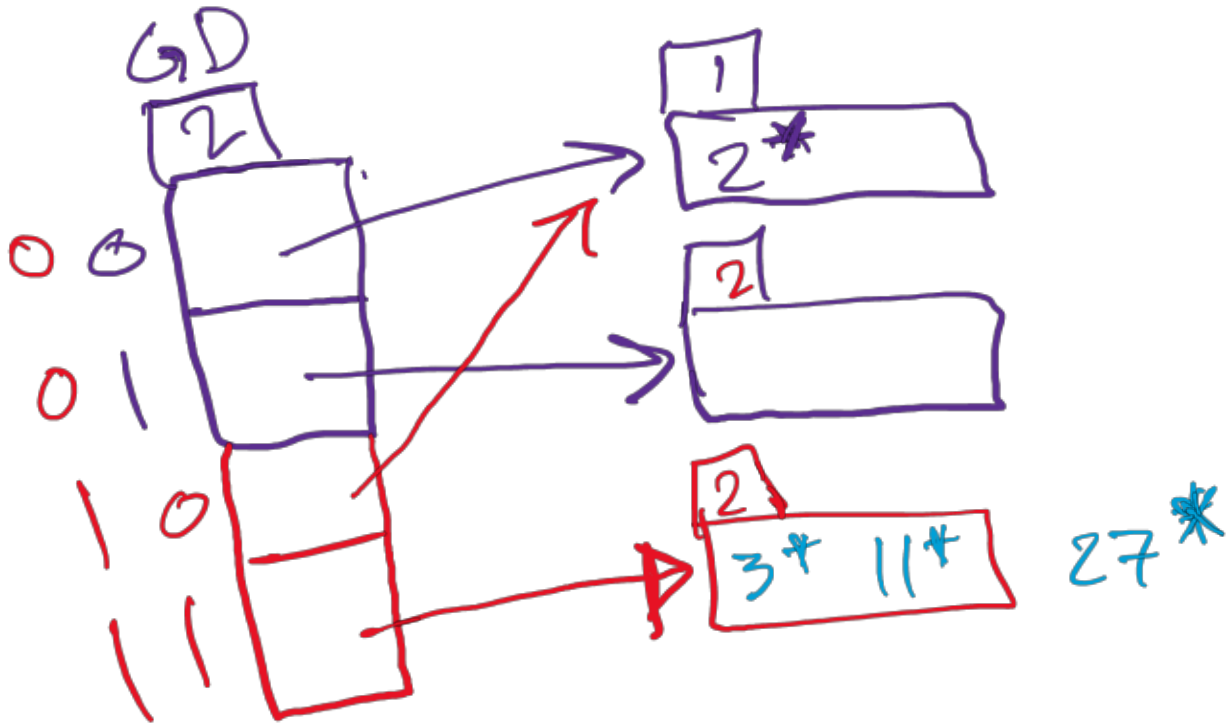

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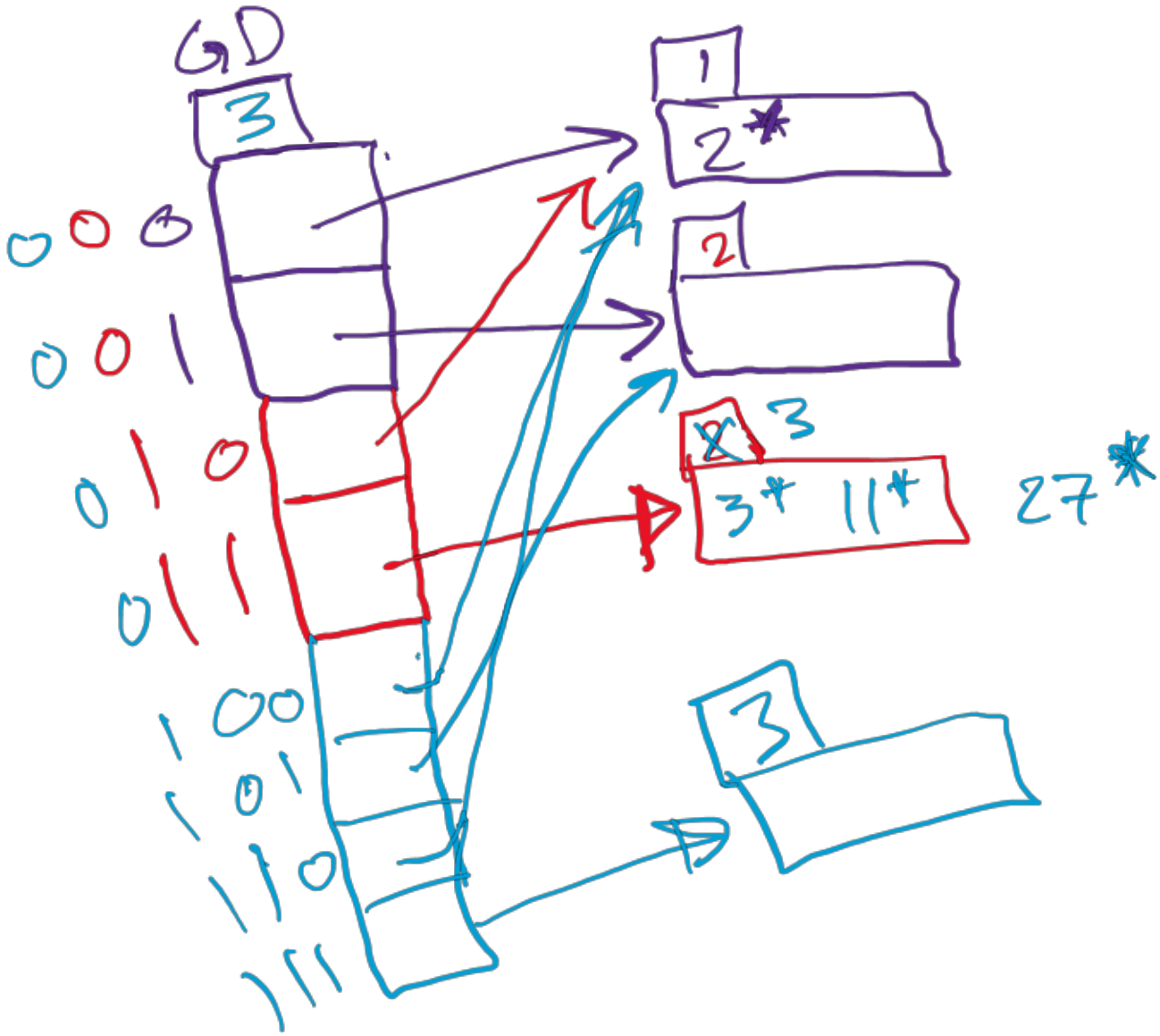

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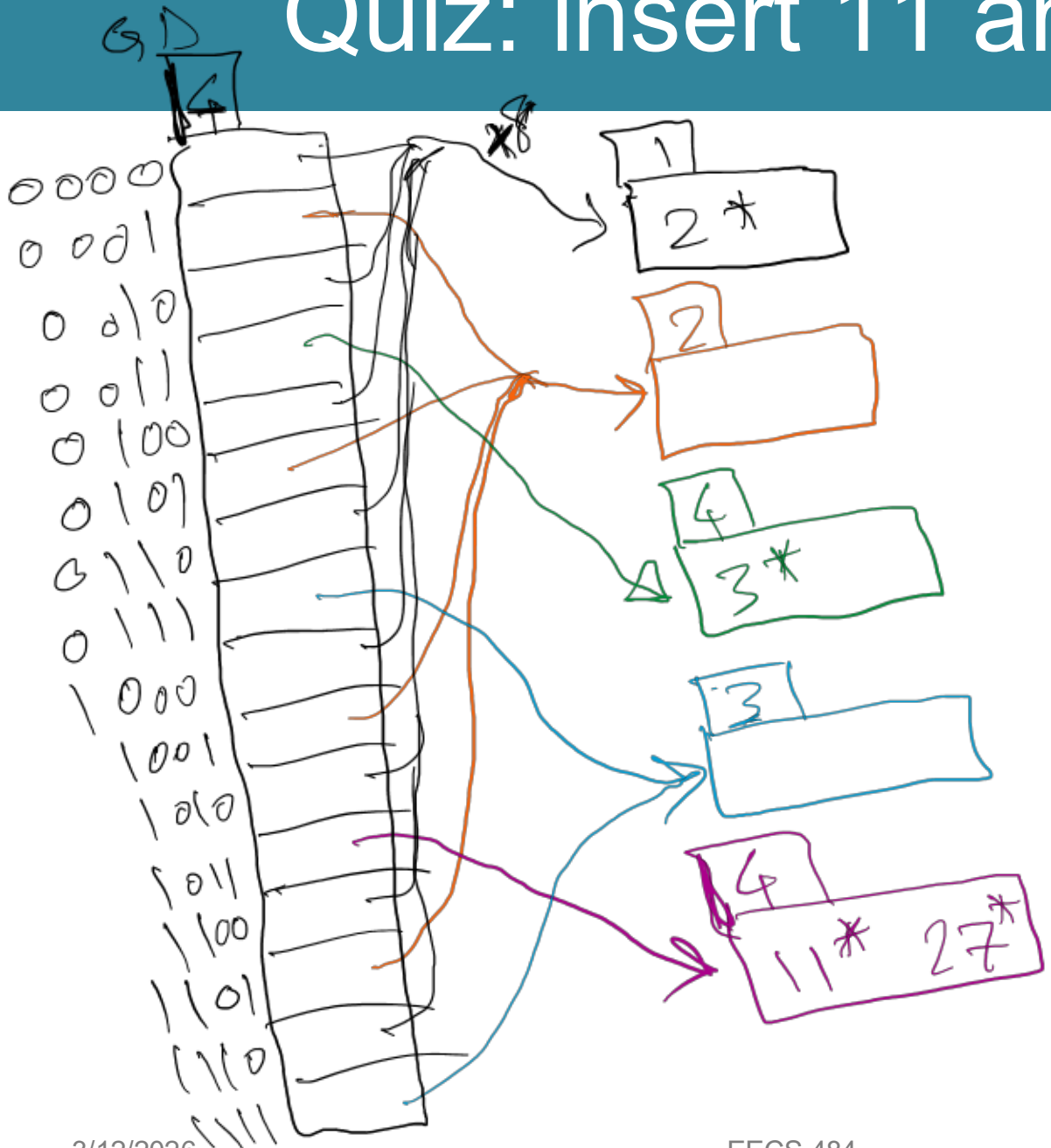

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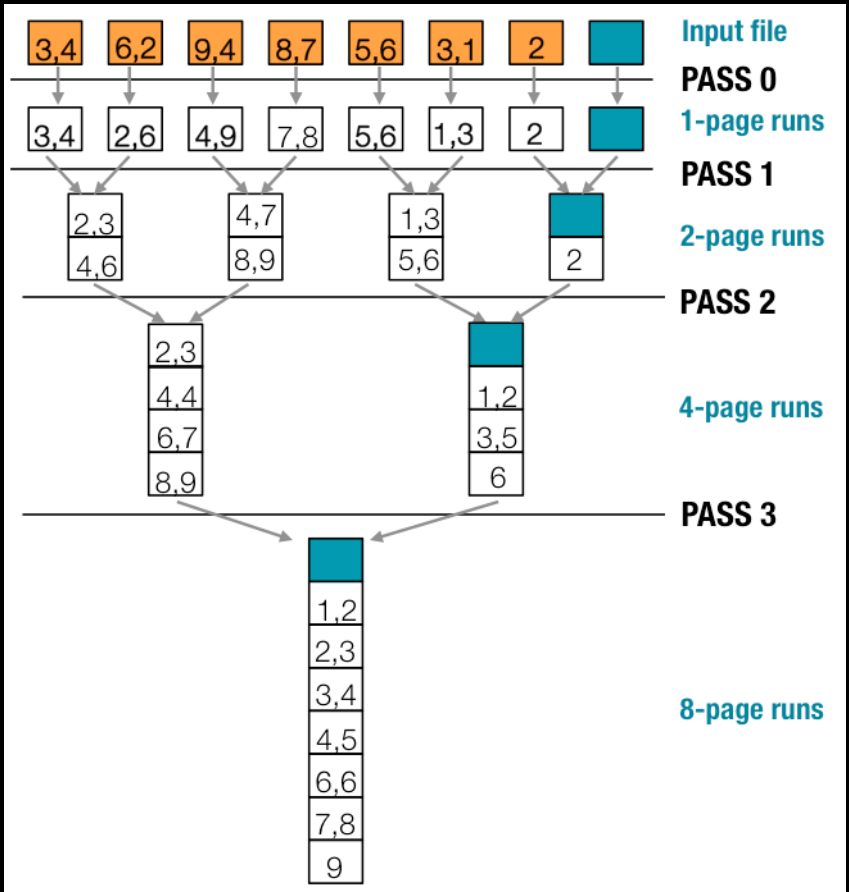
Next Lecture

External Sorting